Conceptualising pedagogical designs for learning through object-oriented collaboration in higher education

Group1

What theoretical concepts are used?

- Sociocultural theory
- Constructivism
- Inquiry-based and project-based learning
  a. Various pedagogical models
  b. e.g., “inquiry-based learning, problem-based learning, and project-based learning are employed to support, in various ways, students’ learning through work on ill-structured, open-ended problems, solved in collaboration and in the context of self-directed activities.”
- Knowledge objects
- Collaborative knowledge creation

What is the significance of theory for the formulation of the aim and research questions in this article?

1. *What were the main characteristics of designing learning with knowledge objects as they emerge based on the two analysed cases?*
2. *How were learning activities with knowledge objects enacted in two higher education courses?*
3. *What were the students' perceived experiences and challenges?*

- Focus on process
- Theory is the fundamental starting point and generate the main goal of the study
- Proven pedagogies (like IBL and PBL) support knowledge creation, they aim to uncover the specific mechanisms of how this collaboration works.
- How practical learning designs in higher education can actually support students' work on shared knowledge objects.

In what way(-s) is this study interested in learning?

- Focus on process of learning through joint creation of editing, revising, etc. a knowledge object with little to no explicit instructions on how to achieve the creation of the knowledge object
- Open-ended
- Learning happens through interactive practices

Is it possible to identify the ontological and epistemological starting point of the study? Why / why not?

Not obvious but implicit in the text

- Learning as a social process that can be observed
- Students actively build up their knowledge by collaborating
- Knowledge objects are evolving entities, shaped by the interactions between participants, and how students learn from engaging in knowledge practice that involves different activities with knowledge objects.

In what ways is theory used to define and describe the research problems in the two studies you have worked with?

Article 1 - DCog states that knowledge is not just inside a person's head, but shared across people, tools, and environments.

Article 2 - sociocultural to define learning as creating shared knowledge objects; shaped by the interactions between participants; Constructivism, learners actively build knowledge and meaning from their experiences and interactions.

In what ways is theory used to frame (conceptualise) the respective problems?

dunno

In what ways is theory used to explore the respective problems empirically?

Article 1: DCog focuses on bodies and environments. The researchers use videos recording of simulator training, and analysis step by step.

Article 2: focus on knowledge objects. Data collection

What are your biggest insights concerning **the use of theories of learning in empirical research?**

Theory plays a role in defining the RQ and how results are perceived. It does not explain the results; it directs entire methodology.